

Meet Parekh

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EDUCATION

International Institute of Information Technology (IIIT)

Hyderabad, Telangana

B.Tech in Computer Science and Engineering; CGPA: 9.17

July 2024 – Present

PROJECTS

Docs++ (Distributed Document System)  | *C, TCP/IP, Sockets* Nov 2025

- **Architected** a concurrent, shared document editing system utilizing a custom TCP-based application protocol.
- **Engineered** a Name Server for load balancing and Storage Servers with data replication to ensure system fault tolerance.
- **Implemented** fine-grained concurrency control (row-level locking) and validated stability with 20+ concurrent clients across 5+ devices.

C-Shell (Custom Unix Shell)  | *C, Linux API* Oct 2025

- **Developed** a custom command-line interpreter supporting 10+ commands, process creation via fork/exec and background process management.
- **Implemented** advanced I/O redirection and multi-stage piping to enable complex chaining of standard outputs and inputs.
- **Integrated** robust signal handling (SIGINT, SIGTSTP) for job control and maintained persistent command history with 15-entry circular buffer storage.

Multi-Threaded Bakery Simulation  | *C, POSIX Threads, Performance Optimization* Nov 2025

- **Reduced** lock contention by **77%** through fine-grained locking and critical section optimization, improving throughput 221%.
- **Engineered** configurable thread pool (1-32 workers) with work-stealing scheduler, achieving 97% CPU utilization.
- **Implemented** performance analytics tracking 15+ concurrency metrics enabling data-driven optimization decisions.

Buy Sell Rent Webapp  | *MERN Stack (MongoDB, Express, React, Node.js)* Feb 2025

- **Built** a full-stack peer-to-peer marketplace application enabling students to buy, sell, and rent items seamlessly.
- **Implemented** secure authentication via JWT and integrated a real-time messaging system for user communication.
- **Designed** a responsive UI with advanced search and category filtering to optimize the user experience across devices.

Aegis Database System | *Python (Flask), MySQL, HTML/CSS* Nov 2025

- **Designed** a highly scalable normalized relational schema enforcing ACID transactions and referential integrity.
- **Implemented** RBAC and parameterized queries to prevent SQL injection vulnerabilities.
- **Developed** complex SQL queries using JOINS and aggregates for a tactical dashboard with real-time data analysis.

SKILLS

Languages: C/C++, Javascript, Python, HTML, CSS

Frameworks: React, Node.js, Flask, Express

Databases: MySQL, MongoDB, Supabase

Operating Systems: Windows, Linux

Misc: Git, Github, Docker, VS Code, Sockets

Libraries: pandas, NumPy, Matplotlib, Manim

ACHIEVEMENTS

JEE Mains 2024: Secured All India Rank 28 (100%ile) among 1.4 million candidates.

JEE Advanced 2024: Secured All India Rank 1263.

Codeforces : Rated 1428 (Specialist); active competitive programmer.

Dean's List 1: Awarded for being in Top 5% of the batch in Spring '25 Semester.

Olympiads: Qualified for Stage 2 in Astronomy (INAO '23) & Chemistry (INCHO '23); NSEC National Top 1% (2024).

POSITIONS OF RESPONSIBILITY

Coordinator, Campus Life Council: Organized 10+ festivals and cultural events on campus.

Corporate Team, The Gaming Club: Managed operational budgets and facilitated professional communication with corporate sponsors.

RELEVANT COURSEWORK

Core CS: Data Structures & Algorithms, Operating Systems & Networks, Data & Applications, Software Systems, Computer Systems Organization

Mathematics: Probability & Statistics, Linear Algebra, Discrete Mathematics